



**AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH**  
 Faculty of Science and Technology  
 Department of Computer Science  
 COE 3206 Computer Networks (Section: ALL)

Final Term Examination

Spring 2022-2023

Total Marks: 100

Moderator: Dr. Md Mehedi Hasan

Time: 2 hours

**General Instructions:**

1. Answer all the questions in the answer script.
2. Do not write anything on the question paper.
3. Return the question paper along with the answer script at the end of the examination.
4. Use pencil / pen to write the answer and to draw diagrams.
5. Marks on the right margin indicate full marks.

**Specific Instructions:**

1. Use of Calculator is STRICTLY prohibited.
2. All the questions are based on OBE program outcomes.

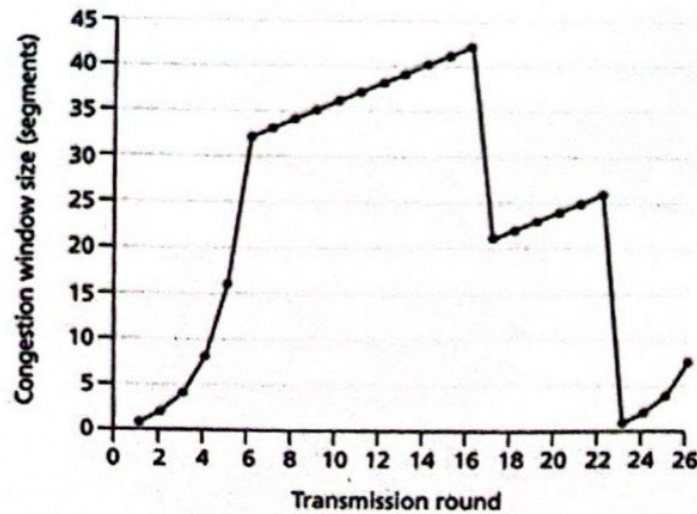
Answer 5 out of 6 Questions

[20 x 5] = 100

1. (a) Detect whether the received sequence 100100011 is error free if the pattern sequence is 1101, using Cyclic Redundancy Check (CRC) code. 8  
 (b) For the parity submatrix P given below, check whether the received sequence 0 1 1 1 0 0 0 is erroneous or not. If it is erroneous, correct the error. 12  

$$p = \begin{matrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{matrix}$$
2. (a) Write the full form of the following ipv6 addresses: 5  
     i) 2001::21:0:0  
     ii) ::2:5:CD  
 (b) Differentiate between unicast and anycast addresses. 5  
 (c) How to find a MAC address from a link local address? Explain considering the following link local address FE80::4E2A:1BFF:FEDF:49E7. 10
3. (a) What is a flat network? Write the advantages of VLAN over the flat network? 5  
 (b) What does VTP stand for? Explain the different VTP modes of operation. 5  
 (c) Which technique enables two end-devices of different VLANs to communicate between themselves? Draw a network to illustrate the concept. 10
4. (a) What is a private network? Explain the necessity of NAT for a private network. 5  
 (b) Write some benefits of NATs/PATs? 5  
 (c) How does NAT make it easier for us to migrate between ISPs? Explain with a proper diagram. 10
5. (a) What is an area in autonomous system? Explain different types of OSPF routers with proper diagram. 5  
     (b) Convert the following ipv4 addresses into ipv6 addresses: 5  
         i) 240.224.15.10  
         ii) 172.92.119.63 10  
 (c) Explain the three-way handshake process of TCP with proper diagram.

6. (a) What is a connection in the light of the transport layer? Write the differences between connection-oriented and connectionless protocols in the transport layer. 5
- (b) Between timeout and 3-duplicate acknowledgment, which one, you think, indicates more severe congestion? Justify your answer. 5
- (c) Consider the following TCP reno scenario: 10



- What is the Threshold of the 1st round [respect to x-axis]?
- What is the Threshold of the 17th round [respect to x-axis]?
- What are the ranges of Congestion Avoidance (CA) in the above figure [respect to x-axis]?
- What are the coordinates (i.e., ranges of rounds) of Slow Start in the above figure [respect to x-axis]?
- What are the reasons of drop in the 17th and 23rd point [respect to x-axis]?